



Probit and Logit Models Transcript

Hello everyone, welcome to lesson 1, video 2 and this lesson we will introduce the probit and logit models. These models are widely used to analyse binary data overcoming the problems that we have in linear probability models. Hi everyone, now in this video we are going to learn about how do we model for binary outcomes, okay.

Now given that we could have a very simple model like this, okay, but the problem with this type of model is you cannot contain the outputs in between 0 and 1, that is what we have already learnt. But what if we transform the line to take a shape like this, S shaped curve like this so that the values, okay, the value of getting S is contained between 0 and 1. So that if it is closer to 1 you already say that there are very high chances of getting S. If it is closer to 0 you say that there are very low chances of getting S, okay. So those are the things which we are going to understand in this case what are the things we use for containing the entire combination or let us say linear combination.

You already see that you have the X, the input data. How do we combine that input data such that the function that we get out of that Z that we get after combining them is actually a function which is contained between or the values are contained between, value of the function is contained between 0 and 1. For that we have two different types of approaches, one the logit model or we call it as logistic function or logistic regression model. The second is the probit model.

This is we generally do not use much but we will more discuss about the logit model because this is one of the things which we generally use, okay. Logistic regression model is actually if you look into the function itself and one of the most peculiar properties of the logistic function is if you think of you have if you take e to the power $\beta_0 + \beta_1 x$ is x , x divided by $1 + x$. So the value itself is contained within 0 and 1. If you think of it, let us say if this particular value, okay, is 0 then the entire value would be 0 because 0 by something is 0 by 1 is 0. This entire value is really really high whatever high value it is. Now you are adding 1 to it.

So basically denominator would be always higher than the numerator hence the value cannot exceed beyond 1. So you have a very good looking formula or a formulation which is called as logistic function which gets you to contain the entire linear combination values because this is the linear combination if you think of $\beta_0 + \beta_1 x$. You are linearly combining your inputs, okay. Once you have the linear combination of your inputs, you can actually make or transform it into a set of values which are in between 0 and 1. They cannot go beyond 1 and cannot be lesser than 0 as well which is what we were looking for. So logistic regression function or logistic function is what we generally use.

We also have the probit model but we will not discuss about it because this is something which we generally do not use. It produces almost identical results which



the logit model gives but let us discuss more about the logit model. So this is how the logit model's function or the logistic function shape looks like, okay.

It is never beyond 0 or lesser than 0, never goes beyond 1 or greater than 1, right. So this is how it looks like but this is the entire intuition, okay. So now you know at this point you have a technique to fit in the values, output values in between 0 and 1 and we cannot use linear regression in this cases, right.

So we have got a method at least at this point which can contain the value of the output or Y values to be contained between 0 and 1, okay. In the next video, we will talk a little more deeper about how do we use all of these things or how do we use logistic function and what is the meaning of that first, okay. In summary, probit and logit models are the standard tools for analysing binary outcomes.

They ensure the valid probabilities are widely used in applied research. In the next video, we will focus more on how do we learn about the coefficients of logit models.

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